

Exploratory Project

B.Tech Civil Engineering

Land Use Land Cover Change in Singrauli District

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Abstract

Land Use Land Cover (LULC) classification is a key application of remote sensing for monitoring the influence of human activity on the physical environment. This study examines LULC change in the Singrauli coalfield region, spanning parts of Singrauli district (Madhya Pradesh) and Sonebhadra district (Uttar Pradesh), over a 15-year span (2006–2021), using multi-temporal Landsat 5 and Landsat 8 imagery. Supervised (maximum-likelihood) classification was performed in ArcGIS 10.0 over a study area of 11,285.56 sq. km, delineating six land cover classes: dense forest, mining areas, settlement/built-up areas, barren land, vegetation and agricultural land, and water bodies. The analysis shows a pronounced decline in dense forest cover (from 17.94% in 2006 to 4.53% in 2021) and in barren land (from 38.71% to 15.25%), accompanied by a marked increase in vegetation and cultivated land (from 23.95% to 52.62%) and in built-up area (from 16.33% to 23.84%). Mining area also grew substantially in relative terms, reflecting the rapid expansion of coal mining and associated overburden dumping. Water body extent declined slightly over the period, attributed to reduced rainfall, siltation from dumpsite runoff, and heavy extraction for thermal power and mining use. The study concludes that the LULC change observed in the Singrauli coalfield is primarily driven by the rapid expansion of coal mining and industrial activity, resulting in significant transformation of the region's fragile ecosystem.

Keywords: Land Use Land Cover; Remote Sensing; Landsat; Supervised Classification; ArcGIS; Singrauli Coalfield; Mining-Induced Land Cover Change

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1 Introduction

Land Use Land Cover (LULC) classification is the process of assigning land cover classes, such as water, urban areas, woodland, buildings, agriculture, grassland, and highland, to pixels in satellite imagery, so that the picture can be naturally grouped into meaningful land cover classes based on the distinctive spectral reflectance of each cover type.

LULC classification of satellite imagery is an important and exclusively studied area within remote sensing; accurate and appropriate land use/cover detection nonetheless remains a challenge. LULC classification plays a noteworthy role in the development of regions and nations: accurate LULC information is vital for urban land planning, agriculture, rural management, and sustainable development. Land use/land cover is an important indicator of global environmental change, reflecting the influence of human activity on the physical environment. Land use/land cover change is a dynamic, widespread, and accelerating process, mainly driven by natural phenomena and anthropogenic activities, which in turn impact the natural ecosystem. Remote sensing technology is extensively used to estimate and monitor such LULC changes, and LULC maps play a significant and prime role in planning, management, and monitoring programmes at local, regional, and national levels.

2 About the Landsat Series

The Landsat series is a joint USGS and NASA-led enterprise for Earth Observation, representing the world's longest-running system of satellites for moderate-resolution optical remote sensing of land, coastal areas, and shallow waters. The first Landsat mission was launched in 1972 as the first Earth Observation satellite with the goal of monitoring the world's land, and was followed by successive missions; Landsat-6 failed to achieve orbit and communication with it was never established, but the series otherwise continues to this day, making Landsat the longest continuous Earth-imaging programme in history.

- **Landsat-1 to 3:** Landsat-1 was launched in 1972 as the first Earth observation satellite with the goal of monitoring the world's land, followed by Landsat-2 in 1975 and Landsat-3 in 1978.
- **Landsat-4 and 5:** Landsat-4 was launched in 1982 and Landsat-5 in 1984, carrying the Multi-Spectral Scanner and Thematic Mapper instruments.
- **Landsat-7:** launched in 1999, carrying the Enhanced Thematic Mapper Plus 8-band whiskbroom scanning radiometer instrument.
- **Landsat-8:** launched in 2013, carrying the Operational Land Imager and Thermal Infrared Sensor instrument.

Data Used

Landsat 8 (for the years 2016 and 2021): Landsat 8 OLI is one of the most widely used remote sensing datasets for LULC classification, providing 11 spectral bands. In this study, bands 1 to 7, namely coastal, blue, green, red, near-infrared (NIR), shortwave infrared I (SWIR1), and shortwave infrared II (SWIR2), were selected for analysis, as bands 8 to 11 are less used in LULC classification.

Landsat 5 (for the years 2006 and 2011): Landsat 5 was a low Earth orbit satellite launched on 1 March 1984 to collect imagery of the Earth's surface. Jointly managed by USGS and NASA as a continuation of the Landsat Program, Landsat 5 carried the same Thematic Mapper and Multi-Spectral Scanner instruments as Landsat 4.

3 Study Area

The study area falls within the geo-coordinates 24°00' to 24°15' N latitude and 82°30' to 82°45' E longitude, lying partly in Singrauli district of Madhya Pradesh and partly in Sonbhadra district of Uttar Pradesh. The area is well connected by motorable road with Varanasi (220 km), Mirzapur (215 km), Rewa (206 km), and Sidhi (130 km). The nearest railway station is Singrauli, on the Chopan–Katni line, which passes parallel to the northern periphery of the coalfield; Renukoot, 45 km from Singrauli, is also nearby. An area of 11,285.56 sq. km was selected as the study area for analysis.

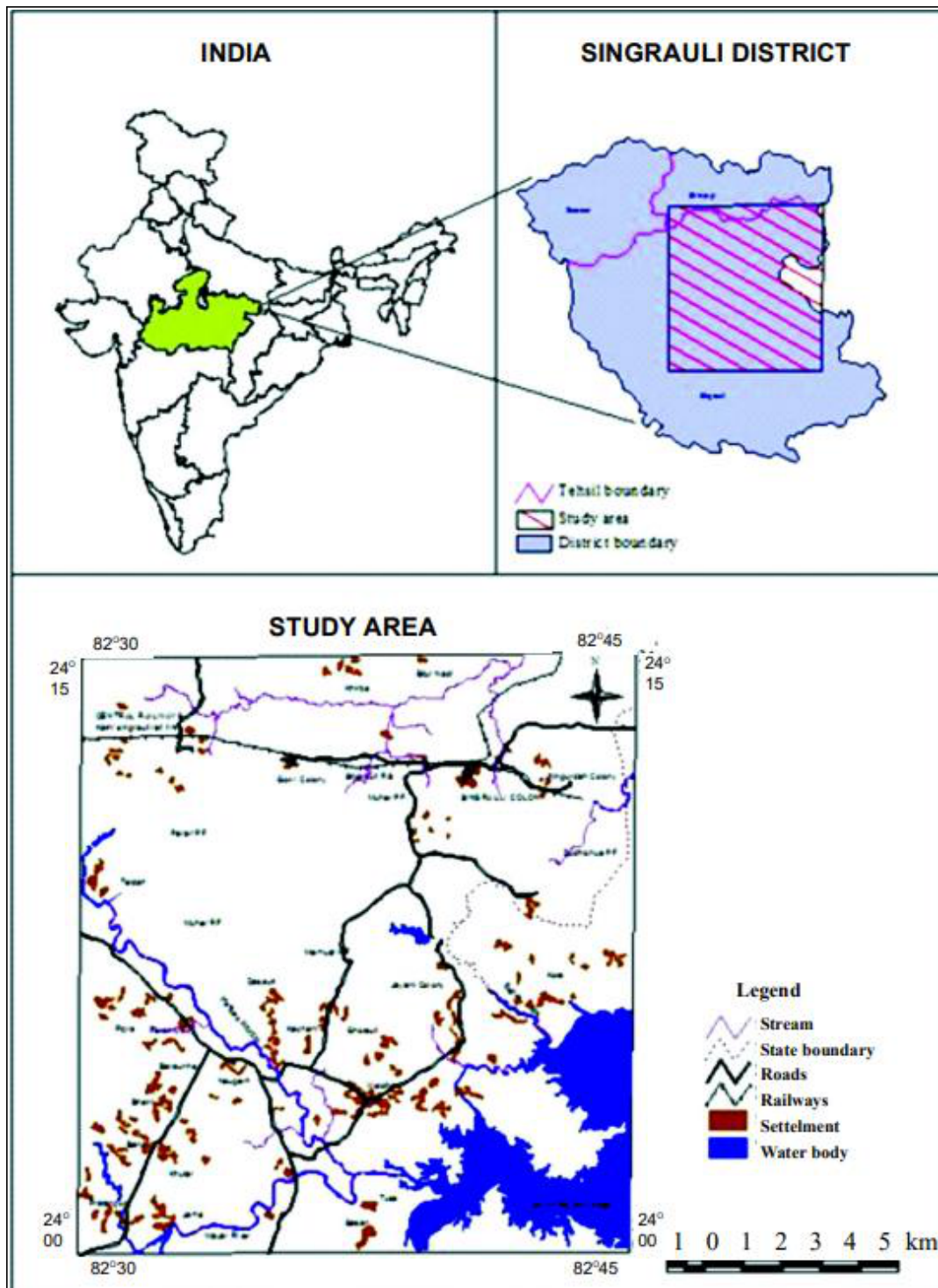


Figure 1: Location map of the study area: India, Singrauli district, and the delineated study area with settlements, streams, roads, railways, and water bodies.

4 Methodology and Application

The classification was performed using **ArcGIS 10.0**, a family of client software, server software, and online GIS services developed and maintained by Esri; ArcMap within ArcGIS 10.0 was used to carry out the classification.

Supervised classification is commonly used for LULC analysis, and ArcGIS provides extensive facilities for standard classification, including tools for classifying numerical fields for graduated symbology. The Interactive Supervised Classification tool was used to accelerate the maximum-likelihood classification process, allowing a quick preview of the classification result without separately running the Maximum Likelihood Classification tool.

Steps of supervised classification:

Composition of bands → creating the shapefile → extraction of the required area → creating training samples → manually classifying pixels using various band combinations → creating the signature file → performing maximum-likelihood classification → converting the final raster layer to a vector polygon → calculating the area.

5 Discussion

The land use/land cover categories delineated in the study area are dense forest, built-up areas, mining areas, barren land, water bodies, and vegetation (including agricultural land).

Dense forest. Dense forest area is deciphered from the False Colour Composite (FCC) satellite image by its dim red tone, coarse-to-medium texture, bordering example, and regular to sporadic shape. Dense forest area covered 17% (2021 sq. km) of the complete region (11,285 sq. km) in 2006, falling to 7% (803 sq. km) in 2011 and 4.53% in 2021, a decline of around 1218 sq. km between 2006 and 2011 alone. Most coal mining activity takes place in the dense forest area, since much of the coal resource lies beneath it; the decrease in dense forest is therefore primarily attributed to the removal of trees as the first step in the expansion of mining activity, removal of forest for infrastructure supporting heavy industrialisation, and increasing population.

Mining areas. Mining pits are interpreted on the FCC image by their black tone, medium-to-smooth texture with a linear to curvilinear pattern, and irregular shape; overburden dumps and thermal power plants can additionally result in loss of agricultural land. Mining occupied 66.49 sq. km (0.5%) of the study area in 2006, increasing to 169 sq. km (1.54%) in 2011, an increase of 102.92 sq. km, as material removed from mining blocks was dumped along the periphery of the plains, forming artificial landscapes. Mining areas generally lack all types of vegetation.

Settlement / built-up areas. Settlement area is situated mostly in the central and south-western regions of the plain area, and is identified on the FCC image by its cyan to light-grey tone. Settlement area grew substantially between 2006 and 2011, an increase of around 1356 sq. km, driven by the development of the industrial sector (NCL and NTPC), which required residential colonies, industrial buildings, schools, and community halls, alongside rising demand for labour that attracted migrants from other states and drove the expansion of villages, towns, and cities in this industrial belt.

Barren land. Rocky, barren area is located mostly on the northern side of the study area, identified on the FCC image by its green-to-brownish tone, coarse texture, and irregular outline. Barren land area changed only slightly during 2006–2016, but its share fell markedly to 15.25%

by 2021, owing to the expansion of settlement and cropland areas.

Vegetation cover and agricultural land. Cultivated land in the study area is under Kharif crops, namely wheat, jowar, masoor, and chana, and is recognised on the FCC image by its red to light-greenish tone, smooth-to-medium texture, non-contiguous pattern, and regular to sub-regular outline. Over the study period, as people returned to and expanded agricultural activity, the vegetation and cultivated land share rose from 23.95% to 52.62%.

Water bodies. Water bodies are identified on the FCC image by their light-blue to black tone, smooth texture, and irregular shape. Water body area fell from 281.12 sq. km (2.4%) in 2006 to 263.05 sq. km (2.33%) in 2010/2011, a decrease of about 18.07 sq. km, attributed to a decline in rainfall of roughly 313 mm between 1978 and 2011, siltation from dumpsite material runoff, and atmospheric dust pollution. The utilisation of water for thermal power plants and mining activity is also an important cause of water body depletion, since usage exceeds the storage capacity of the reservoirs, and washed-out material runoff from dumpsites further blocks natural drainage.

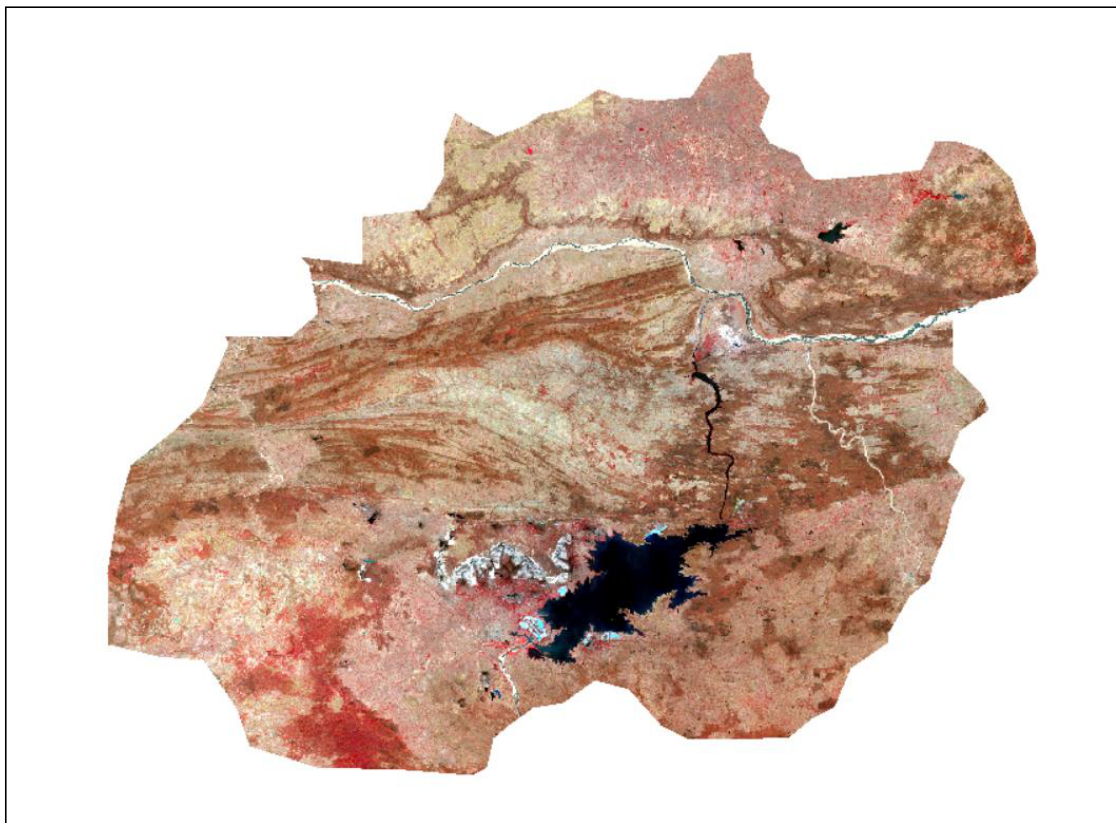


Figure 2: False Colour Composite (FCC) band image of the Singrauli study area.

6 Results

Figures 2–5 present the classified LULC maps of the Singrauli study area for 2006, 2011, 2016, and 2021, and Table 1 summarises the corresponding percentage share of each land cover class over time.

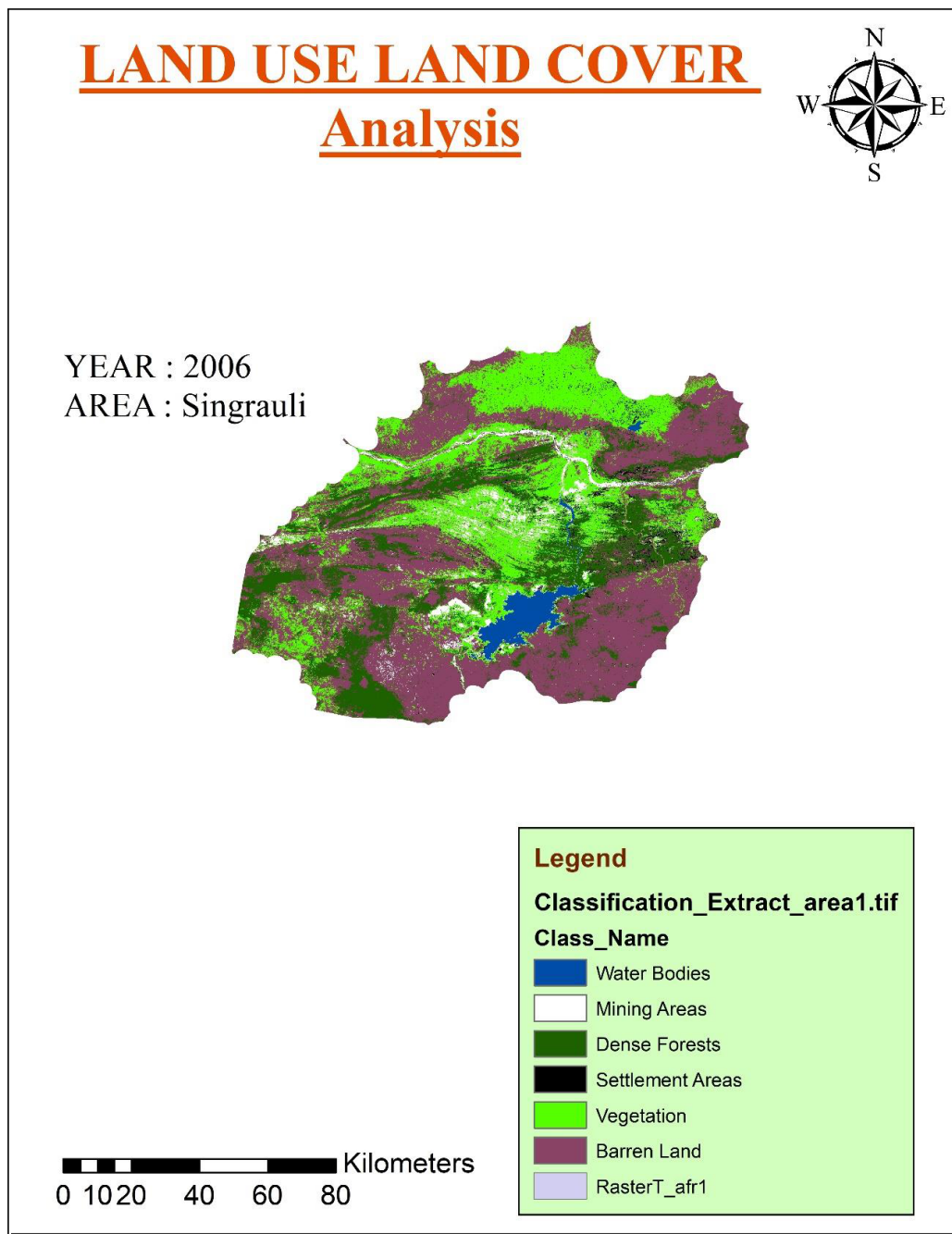


Figure 3: Land use / land cover map of Singrauli, 2006.

LAND USE LAND COVER Analysis

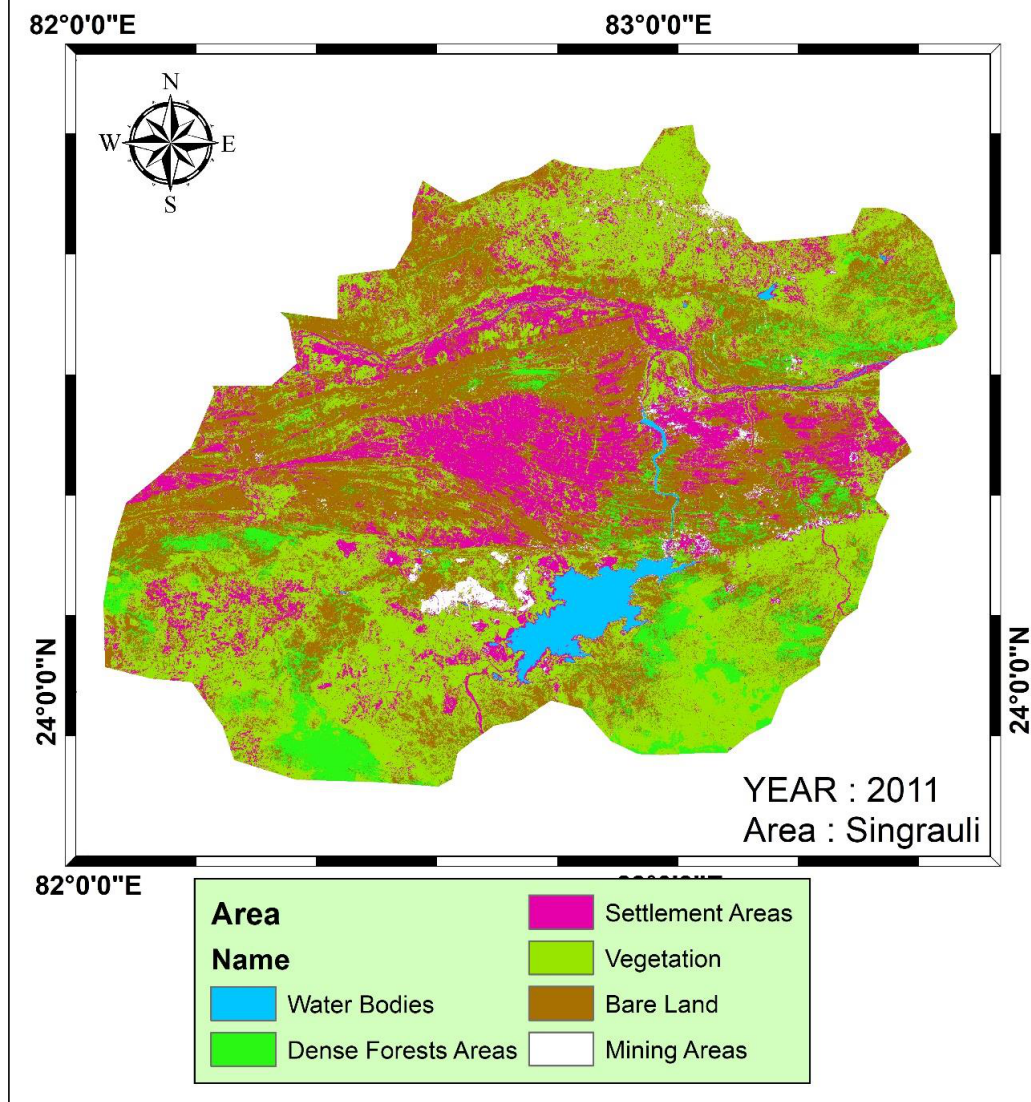
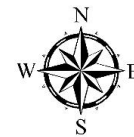
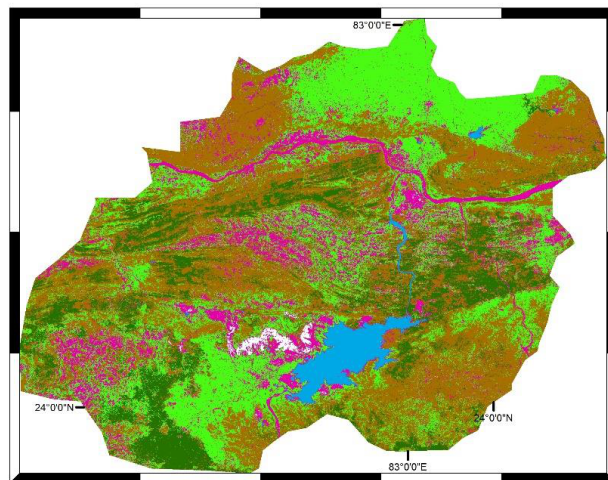


Figure 4: Land use / land cover map of Singrauli, 2011.

LAND USE LAND COVER Analysis

YEAR : 2016

AREA : Singrauli



0 25 50 100
Kilometers

AREA

Name

-  Water Bodies
-  Vegetation + Cultivable Land
-  Built -ip Areas
-  Dense Forest Area
-  Mining Areas
-  Barren Land

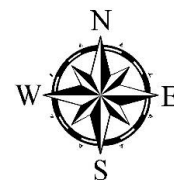
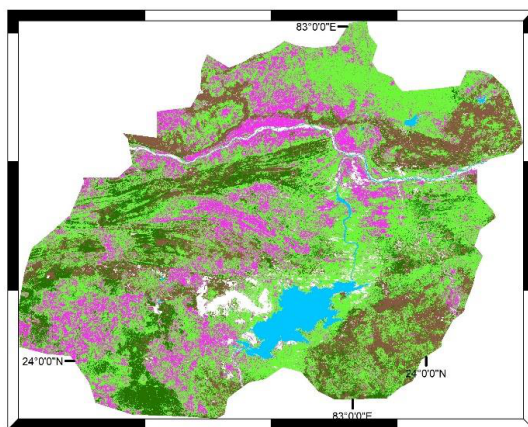
Figure 5: Land use / land cover map of Singrauli, 2016.

LAND USE LAND COVER

Analysis

YEAR : 2021

AREA : Singrauli



Legend

Area

Name

-  Water Bodies
-  Vegetation + Cultivable Land
-  Built -up Areas
-  Dense Areas
-  Barren Land
-  Mining Areas

0 25 50 100 Kilometers

Figure 6: Land use / land cover map of Singrauli, 2021.

Table 1: Percentage distribution of land use / land cover classes, Singrauli study area (2006–2021).

Land Cover Class	2006 (%)	2011 (%)	2016 (%)	2021 (%)
Barren Land	38.71	32.19	32.16	15.25
Vegetation + Cropland	23.95	38.48	38.48	52.62
Dense Forest Areas	17.94	7.12	5.67	4.53
Built-up Areas	16.33	18.43	23.17	23.84
Mining Areas	0.58	1.45	0.21	3.11
Water Bodies	2.49	2.33	0.31	0.65

The distribution confirms the trends discussed above: a steady decline in dense forest and barren land, a corresponding rise in vegetation/cropland and built-up area, and a mining-area share that, while small in absolute percentage terms, grew considerably relative to its 2006 baseline.

7 Conclusion

From this study, it may be concluded that a considerable amount of land cover change has taken place in the Singrauli coalfield over the 15-year span from 2006 to 2021. Large-scale coal mining operations have significantly altered the pre-mining environment; the area under mining shows an increase attributable to the rapid growth in coal production, and while dense forest area is decreasing, plantation efforts at overburden dumps under reclamation schemes have also been ongoing. Overall, the land use/land cover change observed in the Singrauli coalfield has resulted from the rapid expansion of mining and industrial activity during this period, producing drastic changes in the land cover dynamics of this fragile ecosystem.

Acknowledgement

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